

1. (a) State the form in which the following are stored

(i) Unused carbohydrates in plants

(ii) The energy derived from food in humans.

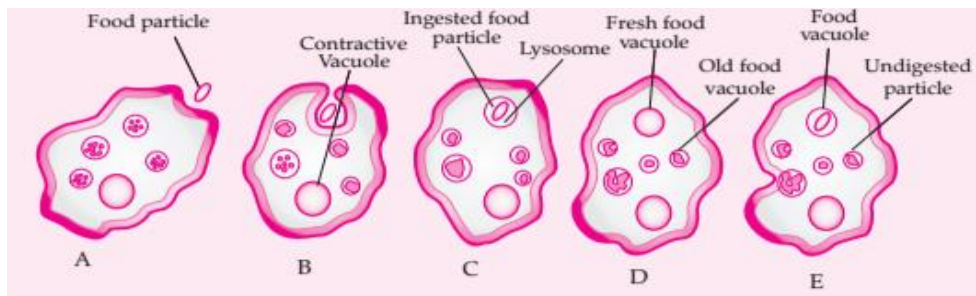
(b) Describe the process of nutrition in Amoeba with the help of diagram.

(c) How does Premium obtain its food?

Answer

(a) (i) Starch (ii) ATP

(b)



Holozoic nutrition in Amoeba

Protozoans like Amoeba capture food with the help of temporary finger-like processes called pseudopodia. As soon as *Amoeba* comes in contact with a food particle or prey, it throws pseudopodia all around the food particle. The tips of encircling pseudopodia fuse and the prey comes to lie in a vesicle or phagosome. This method of intake of food is called circumvallation. Amoeba can also ingest food by other methods like import circumfluence and invagination.

(c) In Paramecium, the food is taken in at a specific spot and is moved to this spot by the movement of cilia, which cover the entire surface of the cell

2. (a) Draw a well labelled diagram of human alimentary canal, and label the following parts:

(i) Liver

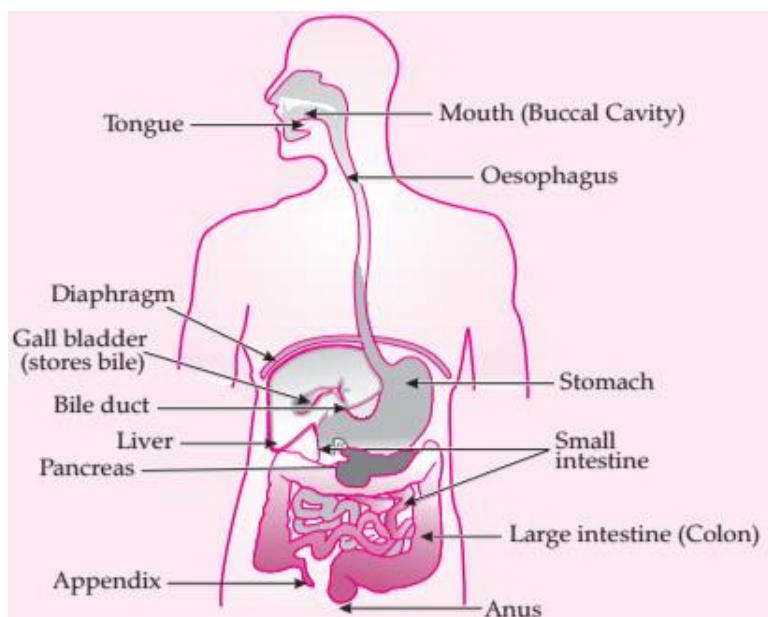
(ii) Pancreas

(iii) Small intestine

(iv) Large intestine.

(b) What is peristaltic movement?

Answer



(b) Peristaltic movement: The lining of alimentary canal has muscles that contract rhythmically in order to push the food forward. This is called peristaltic movements.

3. (i) Name the process and explain the type of nutrition found in green plants. List the raw materials required for this process. Give chemical equation for the mentioned process.

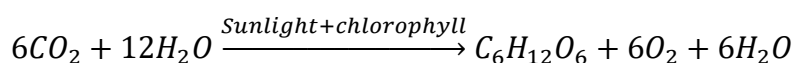
(ii) Write three events that occur during this process.

Answer

(i) Process: Photosynthesis

Type of Autotrophic Nutrition

Explanation: Autotrophic nutrition is the process where plants prepare their own food, using inorganic material such as CO_2 and H_2O in presence of Sunlight and Chlorophyll.



Raw material: CO_2 , H_2O

(ii) (a) Absorption of light energy by chlorophyll.

(b) Conversion of light energy to chemical energy and splitting of water molecules into hydrogen and oxygen.

(c) Reduction of Carbon dioxide to Carbohydrates.

5. Name three different glands associated with the digestive system in humans. Also, name their secretions.

Answer

S. No.	Name of the glands	Name of Secretions
(i)	Salivary glands	Saliva contains enzyme, ptyalin
(ii)	Gastric glands	Secretes gastric juice, HCl, mucus, pepsin.
(iii)	Liver	Bile juice.
(iv)	Intestinal glands	Intestinal juice.
(v)	Pancreas	Pancreatic juice which contains trypsin, lipase and amylase.

Small intestine. Herbivores have longer, small intestine for digestion of cellulose while carnivores have a shorter small intestine for early digestion of meat.